

Sneha Ghosh

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EDUCATION

Pennsylvania State University

M.S. in Computer Science

- Teaching Assistant for AI, Machine Learning, and Data Management for Data Science (100+ students/semester)
- Research Assistant under Dr. Kabir, working on multimodal machine learning for neurological risk stratification

SRM Institute of Science & Technology

B.Tech in Computer Science; GPA: 9.1/10.0; Dean's List; Published at ICIOT'23

University Park, PA

Aug 2024 – May 2026

Chennai, India

Jul 2019 – May 2023

EXPERIENCE

Graduate Research Assistant

Pennsylvania State University

- Developed machine learning models on 10k+ clinical records using Python and PyTorch, building late-fusion 3D ResNet-18 architectures and improving F1-score by 15% through hyperparameter tuning and cross-validation.
- Built and maintained data pipelines across multi-source datasets (imaging and clinical CSVs), applied SHAP and Grad-CAM for interpretability, and led code reviews, reducing analysis cycle time by 20% with zero data leakage.

Artificial Intelligence Intern

GAO Tek Inc.

- Built evaluation frameworks for LLM systems (OpenAI APIs) across 50+ test cases and helped deploy LLM inference pipelines behind REST APIs, improving pipeline reliability by 25% and cutting manual QA overhead by roughly 3 hours/week.
- Worked closely with engineering teams on testing and debugging; findings informed two model iteration cycles adopted across two teams.

AI/ML Engineer

Aeolus Engineering & Technologies

- Wrote clean, efficient Python and C++ to build deep learning models and model-monitoring pipelines for drift detection, cutting mean-time-to-resolution by 40% and reducing processing delays by 20%.
- Contributed to code reviews and mentored teammates, delivering technical solutions across cross-functional engineering teams.

Software Engineer

IDP Global Digital Campus

- Built and maintained backend data pipelines on AWS (S3, Lambda, EC2) processing 1,000+ daily interactions, writing Python and SQL for feature engineering and ML-driven scoring, and improving relevance by 18% while reducing latency by 35%.
- Shipped 5+ production deployments end to end, owning testing, debugging, and reliability improvements alongside senior engineers.

PROJECTS

Agentic Career Copilot | Python, FastAPI, LangGraph, ChromaDB, Docker, GitHub Actions

2026

- Full-stack AI application built end to end: a multi-agent system with a retrieval-augmented (RAG) pipeline, FastAPI backend and REST endpoints, containerized with Docker and deployed via CI/CD – github.com/Snehagh/agentic-career-copilot
- Designed a provider-agnostic LLM layer with pytest coverage and GitHub Actions CI/CD, prioritizing reliability and maintainability over live scraping as a deliberate design trade-off.

Automated ETL Pipeline & Data Quality Framework | Python, SQL, AWS

2025

- Backend data-processing pipeline over the 835-subject ADNI and 1,000-nodule LIDC datasets with automated quality checks and reproducible preprocessing, achieving zero data leakage and a 60% reduction in data prep time. (*Publication in progress*)

Multimodal Lung Nodule Malignancy Prediction | PyTorch, 3D ResNet-18, Grad-CAM

2026

- Deep learning model fusing imaging and clinical features on 1,000 CT scans, reaching AUC 0.97 and 96% accuracy, with Grad-CAM interpretability and embedding clustering (ARI 0.967) to validate learned representations. (*Publication in progress*)

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, Bash

Backend & Cloud: FastAPI, REST APIs, AWS (S3, Lambda, EC2), Docker, Kubernetes, CI/CD (GitHub Actions), Git, pytest

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, XGBoost, HuggingFace, LangChain, LangGraph, RAG, NLP, model evaluation

Data: ETL/ELT pipelines, Pandas, NumPy, feature engineering, data quality, ChromaDB

CERTIFICATIONS & ACHIEVEMENTS

- Certifications: Agent Skills with Anthropic; AI Agents in LangGraph (DeepLearning.AI); Generative AI Fundamentals (AWS)
- Published at ICIOT'23 with two additional publications in progress; Teaching Assistant for 100+ M.S. students at Penn State